MILITARY SPECIFICATION

MANUAL, TECHNICAL: AIRCRAFT STRUCTURAL INTEGRITY PROGRAM - FORCE MANAGEMENT DATA ACQUISITION MAINTENANCE PROCEDURES, PREPARATION OF

"This specification is approved for use by Departments of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense."

1. SCOPE

1.1 This specification covers the detailed requirements for the preparation of a technical manual (-38) that directs special aircraft related maintenance procedures required to support an Aircraft Structural Integrity Program (ASIP). Coverage will provide instructions and guidance to crew members and to personnel in the organizational, intermediate and depot levels of maintenance. Included will be an explanation of the ASIP for a particular series of aircraft.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitations to bids or request for proposal form a part of the specification.

SPECIFICATIONS

Military

MIL-M-38784	Manuals, Technical: General Requirements for Preparation of
MIL-P-38790	Printing Production of Technical Manuals: General Requirements for

STANDARDS

MILITARY

MIL-STD-1530A(11)	Aircraft Structural Integrity Program, Airplane Requirements

PUBLICATIONS

TECHNICAL ORDERS

1-1-1	Cleaning of Aerospace Equipment
1-1-2	Corrosion Prevention and Control for Aerospace Equipment
1-1-3	Preparation, Inspection and Repair of Aircraft Fuel Oil and Water Alcohol Cells and Integral Tanks

TMSS

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1-1-4	Exterior Finishes, Insignia and Markings Applicable to USAF Aircraft
1-1-8	Application of Organic Coatings - Aerospace Equipment
42A31-1-2	General Use of Cements, Sealants, and Coatings
1-1-689	Prevention and Control of Corrosion and Fungus in Communication, Electronic and Meteorological and Avionic Equipment

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 GENERAL REQUIREMENTS

3.1.1 General Specifications: The requirements of MIL-M-38784 and MIL-P-38790 apply as requirements of this specification with the exceptions and additions called out herein. When the general specifications and this specification conflict, this specification shall govern. The technical manual shall be submitted in the form of negatives.

3.1.1.1 Exceptions to MIL-M-38784: The following exceptions to MIL-M-38784 shall apply:

a. Lists. Lists of tables and illustrations are required.

b. Tables. Tables shall not be considered illustrations; they shall be assigned table numbers. Table numbers shall be assigned in the manner specified for figure numbers; for example, Table 2-1 would be the first table in Section II.

c. Blank Pages. Sections shall begin on a right-hand page; portions of sections (subsections) shall not start on a new page.

d. Illustrations. Illustrations may be used more than once, providing such use serves some essential purpose and creates no confusion. Reference in the text to the illustration need not state specifically the purpose of the illustration if the purpose is obvious.

e. Number of Steps in Recorder System Operation, Checkout and Trouble Shooting Procedures. For maintenance procedures, steps shall be assigned consecutive, arabic numerals after each heading. The first step within each subsection shall be numbered "1." If required "Alternate" designation may be used.

f. Reference to other Publications. Reference to other pertinent publications may be made. However, reference to specific paragraphs or sentences shall not be made. If a small amount of information is needed (one printed page maximum) from another publication, the applicable portion shall be extracted, modified if necessary, and included in the manual. If reference to the entire contents or a major portion of another publication is necessary, the cross reference shall be only to the applicable publication number.

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g. Figure Numbers and Figure Titles. Figure numbers and figure titles shall not be used in the maintenance procedures subsections except when required to emphasize a specific procedural step. Figure title may be placed within an illustration.

h. Text Presentation. Text shall be presented in a relaxed format; i.e., single column (typed across the page) with unjustified right hand margin. Final type size of the text shall be approximately 10 point. Art work shall be integrated with the text.

i. Warnings, Cautions, and Notes. Use and format of warnings, cautions, and notes shall be as prescribed in MIL-M-38784.

j. Revisions. Change bars shall be required in revisions.

k. Flagging. Any flagging used shall be explained in the TO introduction.

l. Nomenclature. Nomenclature shall be standardized throughout the manual. If authorized, nomenclature may be shortened or abbreviated to its common name. Standard terminology shall be used in describing recording hardware.

3.2 ARRANGEMENT: The ASIP for each type of aircraft will be peculiar and the manual arrangement should meet the requirements of each individual ASIP. Each individual (aircraft series) manual shall maintain the same arrangement. When a particular section or subsection is not applicable a statement to that effect will be made immediately below the standard section or subsection title. The technical manual shall have the following arrangement:

a. Front Matter

b. Section 1 - ASIP General Description

c. Section 2 - (aircraft designation) ASIP Description

d. Section 3 - (aircraft designation) Aircraft Program Operation

3.2.1 Front Matter: The front matter for each manual shall consist of the following items:

Title Page
List of Effective Pages
Table of Contents
List of Illustrations
List of Relevant General Series Technical Orders and System Peculiar Manuals
Introduction

The Title Page, List of Effective Pages, Table of Contents, and List of Illustrations shall be prepared in accordance with the requirements of MIL-M-38784.

3.2.1.1 List of Effective Pages. The list of effective pages shall be prepared in accordance with the requirements of MIL-M-38784.

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3.2.1.2 Table of Contents. The manual shall contain a table of contents prepared in accordance with the requirements of MIL-M-38784. In addition, each section shall have a table of contents listing the major headings within the section. Text shall begin immediately below the section table of contents. When the table of contents is brief (less than one-half of a page), the Introduction shall begin immediately below the table of contents.

3.2.1.3 Introduction. The introduction shall be prepared in accordance with the requirements of MIL-M-38784. Also the following additional requirements apply:

a. A brief statement of the purpose and scope of the manual and of the content of each section.

b. Definitions of the following terms: Warning, Caution, and Note.

c. Statements concerning technical responsibility for the manual and submission of change requests and comments.

d. Any other data required by the responsible engineering agency.

3.2.2 Section 1 - ASIP General Description. This section shall contain descriptive information about the Aircraft Structural Integrity Program in general. This section shall be broken down into the following subsections:

a. Subsection 1.1 - Program Background. This subsection shall contain ASIP history and general background. The following may be used as a guideline for "Program Background:" Following the investigation of the B-47 accidents in late 1950's, HQ USAF directed the Aeronautical Research and Development Center, (now ASD), at WPAFB to take all necessary steps to ensure an adequate fatigue or service life of first-line aircraft. Historically, new aircraft were subjected to extensive engineering analysis and laboratory testing to make sure the structural strength of the aircraft meets the needs of the operating command. Evaluating the service life of an aircraft was a new concept that required the engineers to develop analysis and testing programs tailored specifically to fatigue life validation. The HQ USAF directed program of certifying both the structural strength and fatigue life of an aircraft is the basis of todays Aircraft Structural Integrity Program.

b. Subsection 1.2 - Program Objectives. This subsection shall contain a discussion of MIL-STD-1530A(11) and AFR 80-13 as it relates to this manual.

c. Subsection 1.3 - Tasks. This subsection shall contain a brief description of Task I (as described in MIL-STD-1530A(11)) through Task III and a more detailed description of Task IV and V of the Force Management Program.

d. Subsection 1.4 - Interaction with Other Programs. This subsection shall describe the integral part the operational aircraft programs play in the overall ASIP.

e. Subsection 1.5 - Applicable References. This subsection shall list all references required to support the Flight Recorder and Individual Airplane Tracking (IAT) programs. These references include: applicable handbooks, specifications, standards, technical orders, regulations, instructions and work unit codes.

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3.2.3 Section 2 - (aircraft designation) ASIP Description. This section shall contain descriptive information about the (aircraft designation) ASIP. This section shall be broken down into the following subsections:

a. Subsection 2.1 - Force Management (Task V) Operational Programs. This subsection shall contain a general description of the applicable force management (Task V) Operational Programs for the aircraft. The Flight Loads Recorder Program shall be listed and described as an integral part of the Loads/ Environment Spectra Survey Program. The following operational programs shall be listed and described as an integral part of the Individual Airplane Tracking (IAT) Data Program:

Exceedance Counter Program
Mechanical Strain Recorder Program
IAT Forms Program (to include AFTO Form 238 for component serialization report)

b. Subsection 2.2 - Data Flow. This subsection shall contain graphical flow diagrams of flight loads recorder data through the Loads/Environment Spectra Survey Program and exceedance counter, mechanical strain recorder and individual airplane tracking (IAT) data through the IAT Program. Further, flow diagrams shall show the information feedback to organizational, intermediate and depot levels of maintenance. A brief discussion of each task/step in the graphical flow diagram shall also be included.

c. Subsection 2.3 - Functional/Management Responsibilities. This subsection shall contain detail instructions for recorder or forms management reporting/information collection responsibilities. Single points of contact for each functional/management area shall also be included in this subsection.

3.2.4 Section 3 - (aircraft designation) Aircraft Program Operation. This section shall contain either reference to existing aircraft TOs or detailed system description, operation, checkout, maintenance/trouble shooting and data forms instructions for the following flight data gathering programs (if applicable to aircraft series):

a. Flight Loads Recorder Program

b. Exceedance Counter Program

c. Mechanical Strain Recorder Program

d. Individual Airplane Tracking (IAT) Forms Program

3.2.5 Alphabetic Index. An alphabetical index shall be prepared in accordance with the requirements of MIL-M-38784.

3.3 MAINTENANCE. Maintenance of the manual shall be in accordance with MIL-M-38784, with the exception called out in 3.1.1.1 herein.

3.4 NEGATIVES. Negatives furnished shall meet the requirements of MIL-M-38784.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Manuals (Basic and Revisions)

4.1.1 In-process Review. Draft manuscript shall be reviewed as scheduled during the writing process. Comments and corrections resulting from this review shall be incorporated into the review copy of preliminary technical order (TO). The provisions of TO 00-5-1 for validation shall apply.

4.1.2 Review Conference Verification. Unless otherwise specified in the contract, a review conference shall be held at the contractor's facility, or if combined with a verification, at an operational location. Review copies shall normally be in the form of a preliminary TO, with text double spaced and containing all required art. The manual shall be reviewed/verified for technical adequacy and accuracy. Classification markings, if required, shall also be reviewed at this time. Before preparation of reproducible copy is begun, a final manuscript review may be required to insure incorporation of all review/verification inputs.

4.1.3 Prepublication Review. Reproducible copy of the manual shall be reviewed by the Government as contractually specified for technical adequacy and accuracy and for conformance to specifications. Reproducible copy shall meet the requirements of MIL-M-38784.

4.1.4 Review of Reproducible Copy. Prior to preparation of negatives, and after the contractor quality control review, reproducible copy shall be reviewed by the responsible engineering agency for technical accuracy and conformance to specifications.

4.1.5 Proof Copies. Four proof copies, reproduced from the final negatives by ozalid or other means, shall be forwarded to the responsible engineering agency on or before the date of shipment of negatives.

4.2 CHANGES

4.2.1 Preliminary Review. Preliminary draft of the manuscript shall be submitted to the responsible engineering agency for review. Manuscript for changes shall be clear and legible and shall include as a minimum, a listing of the art work to be used. Comments and corrections resulting from this review shall be returned to the contractor for inclusion in the final negatives.

4.2.1.1 Review of Reproducible Copy and Proof Copies. Reproducible copy of changes shall be reviewed and proof copies furnished as specified for the basic manual.

4.3 REVISIONS. Revisions shall be prepared, reviewed and approved, and proof copies furnished as specified for the basic manual.

5. PREPARATION FOR DELIVERY

5.1 NEGATIVES. Negatives shall be prepared for shipment as required in MIL-P-38790. The provisions of MIL-P-38790 relative to use of reproduction assembly sheets with negative shipments shall apply.

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6. NOTES

6.1 Intended Use. Technical manuals prepared to this specification are for use of maintenance personnel for support of an Aircraft Structural Integrity Program (ASIP) and associated Force Management Data Acquisition Programs (flight data gathering).

6.2 Responsible Engineering Agency. The Air Force ASIP OPR (as defined by AFR 80-13 AFSC/AFLC Supplement 1) is the responsible engineering agency for technical manuals prepared according to this specification.

6.3 Review of Reproducible Copy. The review of reproducible copy is not intended to replace the contractor's quality control inspection.

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